main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>
#include <ctype.h>

char input[100];
int i = 0;

int E();
int EP();
int T();
int TP();
int F();

int E()
{
    if (T())
    {
        if (EP())
            return 1;
        else
            return 0;
    }
    return 0;
}
```

Output

```
Recursive Descent Parsing
Grammar:
E  -> TE'
E' -> +TE' / epsilon
T  -> FT'
T' -> *FT' / epsilon
F  -> (E) / id

Enter the string to be checked: a+b*c

String is accepted
```

main.c

Run ▶

```
int EP()
{
    if (input[i] == '+')
    {
        i++;
        if (T())
        {
            if (EP())
                return 1;
            else
                return 0;
        }
        return 0;
    }
    return 1;
}

int T()
{
    if (F())
    {
        if (TP())
            return 1;
        else
            return 0;
    }
```

Output

```
Recursive Descent Parsing
Grammar:
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Run ▶

```
52      return 0;
53  }
54
55  int TP()
56  {
57      if (input[i] == '*')
58      {
59          i++;
60          if (F())
61          {
62              if (TP())
63                  return 1;
64              else
65                  return 0;
66          }
67          return 0;
68      }
69      return 1;
70  }
71
72  int F()
73  {
74      if (input[i] == '(')
75      {
76          i++;
```

Output

```
Recursive Descent Parsing
Grammar:
E  -> TE'
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main.c

Run ▶

```
        if (E())
        {
            if (input[i] == ')')
            {
                i++;
                return 1;
            }
            return 0;
        }
        return 0;
    }
    else if (isalpha((unsigned char)input[i]))
    {
        i++;
        return 1;
    }

    return 0;
}

int main()
{
    printf("Recursive Descent Parsing\n");
    printf("Grammar:\n");
    printf("E  -> TE'\n");
```

Output

```
Recursive Descent Parsing
Grammar:
E  -> TE'
E' -> +TE' / epsilon
T  -> FT'
T' -> *FT' / epsilon
F  -> (E) / id

Enter the string to be checked: a+b*c

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```

main.c

Run ▶

```c
int main()
{
    printf("Recursive Descent Parsing\n");
    printf("Grammar:\n");
    printf("E  -> TE'\n");
    printf("E' -> +TE' / epsilon\n");
    printf("T  -> FT'\n");
    printf("T' -> *FT' / epsilon\n");
    printf("F  -> (E) / id\n");

    printf("\nEnter the string to be checked: ");

    if (fgets(input, sizeof(input), stdin) == NULL)
        return 1;

    input[strcspn(input, "\n")] = '\0';

    i = 0;

    if (E() && input[i] == '\0')
        printf("\nString is accepted\n");
    else
        printf("\nString is not accepted\n");

    return 0;
}
```

Output

```
Recursive Descent Parsing
Grammar:
E  -> TE'
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T' -> *FT' / epsilon
F  -> (E) / id

Enter the string to be checked: a+b*c

String is accepted
```